information in the cell is called the *genom*. Stretched out the total genetic material is about 1.5 m long, but its diameter is only 2 nm; the molecular weight is about 10^{11} per chromosome. While the cells comprise most of the body, the volume of the chromosomes only occupy ~1% of the cell volume and only about 1/10 of the chromosome volume carries a genetic message; thus the genetically significant target is rather small. Thus the probability that radiation should directly "hit" the DNA is small compared to the probability that radiolytically produced radicals will react chemically with the chromosomes.

The nucleotides are held together by hydrogen bonds between the nitrogen bases, which have their nitrogen rings perpendicular to the plane of Figure 15.1.I (i.e. turned 90° compared to the paper surface). The nitrogen rings contain some π-electrons, which interact between the planes and stabilize the chain structure. The unsaturated bonds in the nitrogen bases are sensitive to oxidation. Further, the phosphate oxygens are ionic. Thus, the DNA presents all types of chemical bonding and is sensitive to many types of reactants.

The *cell cycle* plays an important role in radiation damage (Figures 15.1.B and C) (i) during the S-stage proteins are synthesized, (ii) in the "gap 2", G_2, stage, the cell is being prepared for division (the DNA-chains are split up and copied), (iii) in the *cell division* (*mitosis*) the chromatin condenses into chromosomes, which are split and move in opposite directions, while the original cell is divided at mid-level (Figs. 15.1.C.3 and C.4), after which (iv) the cell "rests" or matures in the G_1-state.

15.2. RADIATION EFFECTS ON THE MOLECULAR LEVEL

When a high energy particle (whether α, β or γ) strikes a human cell it produces a narrow track, less than 1μm thick (cf. Figs. 7.5 and 8.1). The interaction with water, which is the main constituent of the cell, leads to radiolysis products, as described in Chapter 8. To a lesser extent, the radiation may interact directly with the molecules of the cell ("direct hit"). Only the effects (direct or indirect) on the cell nucleus DNA are thought to cause severe biological damage.

15.2.1. Radiation physics

High energy γ-rays lose most of their energy in a few compton scattering events. A "typical" natural radiation background gamma quantum of 1 MeV loses about 75% of its energy in 20 cm of water (a "typical" human thickness). This loss occurs to 50−80% in a single compton scattering event, producing an electron with up to 0.8 MeV kinetic energy, depending on the scattering angle. In the next compton interaction, the ≥0.2 MeV γ loses ≥50% of its energy; the distance between these initial two interactions exceeds cellular dimensions. Thus, the 1 MeV γ-absorption on the cellular level can be considered as the production of a single high energy electron. However, at very low

energies (compton γ's and X-rays), low energy electrons are produced at densities of several electron pairs per μm (cf. Fig. 7.5.d and Table 7.2; 1 mm air is considered equivalent to about 1 μm of water or tissue).

High energy electrons lose most of their energy in low energy collisions (cf. Figures 7.5 and 7.7). A typical LET-value is 200 eV/μm (Table 7.2), corresponding to 5−10 ion pairs/μm; a 10 keV electron produces about 10 ion pairs/μm according to Figure 7.7.

Alpha particles cause dense ionization (Figs. 7.5 and 7.7) with typical LET-values of 200 keV/μm (Table 7.2) and production of several thousand ion pairs per μm (i.e. several ion pairs/nm).

High energy neutrons are absorbed in water mainly by collisions with H-atoms, forming energetic protons which ionize similar to alpha particles (Table 7.2 and Fig. 7.7). In the collision process the H-atoms are knocked from their positions in the molecule; also other atoms may be knocked out. The neutron ends its life by being absorbed in a (n,γ)-process or leaves the system.

15.2.2. Radiation chemistry

The absorbtion of low-LET radiation and high-LET radiation is illustrated in Figure 15.1.E and 15.1.H, respectively. Table 15.2 gives the number of ion pairs formed in the DNA, nucleosome and chromatin by α's and γ's (^{60}Co). Table 15.3 shows the damage caused to the DNA and cell nucleus. The wide range of values in parentheses indicate the uncertainty as well as the dependence on the particle energy (in general, higher ionization at lower energy).

The radiation is largely absorbed in water, as the cell contains some 70% H_2O, and produces ions, free radicals and excited atoms. A momentaneous lethal dose of 20 Sv

Table 15.2 Ionization clusters produced directly in a DNA-related target by a single radiation track (From Goodhead, UNSCEAR)[†]

	Average number of ionizations in		
Radiation	**DNA segment (2 nm long)**	**Nucleosome (∅ ~ 10 nm)**	**Chromatin segment (25 nm long)**
γ-rays	1 (0 to <8)	2 (0 to <20)	2 (0 to <45)
α-particles	2 (0 to <15)	10 (0 to <90)	50 (0 to <200)

[†]UNSCEAR = United Nations Scientific Committee on the Effects of Atomic Radiation.

Table 15.3 Damage products in a single-cell nucleus traversed by a single radiation track (From Goodhead, UNSCEAR)

		Average number of induced breaks	
Radiation	**Average number of ionizations**	**DNA single strand**	**DNA double strand**
γ-rays	70 (1−1500)	1 (0−20)	0.04 (0−few)
α-particles	23 000 (1−100 000)	200 (0−400)	35 (0−100)

produces a concentration of 14 μM of reactive products ($\cdot OH$, e_{aq}^-, H_2O_2, etc). Trace metals in the body (e.g. Cu, Cr, Se) are also poisonous at this concentration level, but essential at lower levels (e.g. a daily intake of 40 μg Se is recommended). The ions produced will probably have little effect as the DNA contains numerous ionizable positions at the phosphate group. Free radicals like $HO\cdot$ and oxidizing products like H_2O_2 are highly reactive and can add to unsaturated bonds, which upsets the sensitive hydrogen-π-bonding and may break the bonding between the two helices. Excited hydrolysis products may transfer the excitation energy to the DNA, leading to a localized break in the sugar-phosphate chain. The damage to the DNA may also lead to a substitution reaction in the nucleotide or a loss of a segment. Actually, hundreds of different DNA-damage products have been identified. If the damage is limited to one of the strands of the helix, it is referred to as a *single strand break*. These damages occur naturally due to what is called oxidative metabolism at a rate of about: 3000 single strand breaks/cell/day, 3000 base changes/cell/day and 8 double strand breaks/cell/day.

The cell is protected by different *DNA repair mechanisms*, which all try to restore the damage. This repair typically result in one of several results. The most common one is an error free repair. However, there are also the possibility of no repair at all which lead to mitotic death of the cell. Even if the repair is made it can be erroneous and this can have two results. Either the cell suffer apoptosis which is the process of programmed cell death, or the evolution of cancerous behaviour, or an inheritable DNA-defect. The details are not known, except when the repair goes wrong (e.g. a replacement of a lost nucleotide by a "wrong" base-pair, etc.). It is believed that most single strand breaks are correctly repaired. The repair system is believed to be more effective in a living organism, where the cells are in continuous exchange with surrounding cells and body fluids, than in the tissue samples often studied in the laboratory. This should be kept in mind below, where effects on whole organisms as well as on cell cultures in "test tubes" (*in vitro* studies) are described.

The cell contains natural *radical scavengers*. A radical scavenger is a molecule which reacts readily with the radicals or other oxidising agents why they are also sometimes referred to as anti-oxidants in biological systems. As long as the radical scavangers are in excess of the radiolysis products, the DNA may be protected. When the products exceed the amount of scavengers, radiation damage and cancer induction may occur. In principle, there could thus be a *threshold* dose for radiation damage, at which the free radicals formed exceed the capacity of scavenging. The scavenging capacity may differ from individual to individual, depending on his/her physical condition.

A low-LET gamma ray "hit" may cause 1 ionization (or <8) in a 2 nm DNA-segment. This corresponds to a deposition of some 30 eV (or <250 eV) in about 6 base-pairs + sugar-phosphates. The radiation energy may distribute over a large number of bonds, so that no bond gets enough energy to dissociate. Thus, the radiation interaction may leave the DNA-segment mainly unchanged.

Low-LET radiation sometimes form clusters of ions along the particle track, i.e. produces high-LET "spots". Such spots mean increased risk of damage because a larger amount of energy is then deposited in the small volume occupied by the DNA-helix. High-LET spots therefore increase the possibility of damage to both strands of the helix, causing a *double strand break*.

It has been found that 1 mGy leads to 1 single strand break in each cell, 1 double strand break in every 25 cells and one aberration in each 10000 cells. Because the DNA-chain is rather rigid, a double strand break does not lead to the two halves snapping away from each other. Instead, they are resting, waiting for the repair mechanism to start. Nevertheless, the repair is more difficult and the chances of "repair errors" (mutations) are much larger than for the single strand break. This is demonstrated by two facts: (i) the repair time for a single break is on average 10 minutes, while that for the double break some hours; (ii) *chromosome abberations* (Fig. 15.1.D) only occur after double strand breaks. The chromosome abberations may be due to that the cell undergoes mitosis before the double strand break has been repaired. Cells with severe chromosome abberations are not viable. Minor repair errors may be carried on to next generation of cells as a change in the genetic code or may stop the cell division process.

In addition to double strand breaks being more difficult to repair than single strand breaks, double strand breaks caused by high-LET radiation are more difficult to repair than those caused by low-LET radiation. The high energy deposited at the hit area causes "havoc" on the DNA (cf. Table 15.3). A few percent of low-LET radiation causes double-strand breaks in contrast to 10–20% for high-LET radiation.

After a low-LET dose of 3 Gy, every cell in the human body contains 3000 single strand and some 100 double strand breaks above normal; if given in a short time the cell (and organism) has great difficulties in repairing this damage (resulting in death in the case of no medical treatment). However, spread out over a week only chromosomal abberations will be noted. In the subsequent sections, we shall discuss macroscopic effects of various kinds and doses of radiation.

One effect of the time needed for repair of damages cells is that radiation sensitivity is larger the faster the cells divide (the Bergonié-Tribondeau law). Hence cancer cells are normally more sensitive to radiation than slower dividing normal cells of the same kind. This is one of the reasons why radiation is used in cancer therapy.

15.2.3. Radiation weighing factors

In order to take into account the biological effects of different kinds of radiation, *radiation weighing factors*, w_R, have been introduced and are given in Table 15.4. Earlier, two similar concepts were used with about the same meaning: the "quality factor", Q, and "relative

Table 15.4 Radiation weighting factors, w_R (ICRP 2007)

LET in water (eV/nm)	Weighting factor w_R	Type and energy of radiation
0.2–35	1	photons (X-rays and γ's)
	1	electrons, muons
	2	protons and charged pions
130	20	α-particles, fission fragments, heavy ions
		continuous function of n-energy neutrons

biological effectiveness", RBE. The ICRP 1990 dose concept, the biologically effective dose, is termed *equivalent dose* and abbreviated H_T, and defined as:

$$H_T\,(\mathrm{Sv}) = \sum_R w_R D_{T,R} \tag{15.1}$$

$D_{T,R}$ is the absorbed dose averaged over the tissue organ T, due to radiation R. H_T is measured in sieverts (Sv); 1 Sv = 1 J kg^{-1}. The earlier, but similar concept was the *rem* (radiation equivalent man), where 1 Sv equaled 100 rem. Until 1990, H_T was called dose quivalent. $D_{T,R}$ is measured in J/kg, which in this context is called gray (Gy). The summation is taken over all sources irradiating the target.

15.3. RADIATION EFFECTS ON DIFFERENT TYPES OF CELLS

We can distinguish between two types of cells: those which are directly involved in the functioning of the organ (e.g. the cells of bone marrow, liver, or the nervous system) and those which are associated with reproduction. Radiation damage gives rise in the former to *somatic effects* (i.e. limited to the organism irradiated) such as cancer induction or cell death, and in the latter to *genetic effects* (i.e. limited to future generations).

Cells which are undergoing frequent division, and organs and tissues in which the cells are replaced slowly, exhibit high radiation sensitivity. Of the some 200 different kinds of cells in our body, some never divide (e.g. in the ovary, some sense organs and part of the central nervous system, except in the embryo state), while others divide frequently (bone marrow, intestinal epithelium, male gonads). The cell cycle time varies from hours to days. Usually, tumor cells divide much faster (3–5 times) than surrounding healthy tissue.

In general, the more differentiated the cells of an organ (i.e. the higher the organ is on the biological evolutionary scale), the greater the sensitivity to radiation. This also holds for different organisms, as reflected in Figure 15.2 and Table 15.5. Figure 15.2 shows the *inactivation dose* (i.e. leading to no further cell division) for organisms with different cell